

Lab Week 6

Using Oracle data modeler to construct a logical data model using the IE notation to fit the following circumstances.

Draw data models for the following situations. In each case show also all cardinalities.

- a) A School will employ one or more faculty members, but a faculty member can only belong to one School. For each school retain the unique school id and the school name. For each faculty retain the unique faculty id, the faculty name, rank, current salary and their start date. A faculty member must belong to a School
- b) RU has many majors. Each major can have many students enrolled but a student can only have 1 major. For each major retain the unique major abbreviation, the full major name and its brief description. For each student retain their student id, their name and birth date. However not all students have a major
- c) The Bay Crest hospital retains data on its beds and their occupancy. Each bed is identified by a combination of the floor level and room number where it is located, and a bed number assigned uniquely to each bed within a particular room. The type of bed is also retained (e.g. semi-private, private, intensive care, etc.). Information is also retained about patients, both those assigned to a bed as well as outpatients. This consists of a unique patient id as well as the patient name, address, gender and date of birth. At any one time some beds are not occupied.
- d) The TSO orchestra employs many musicians. Each musician plays only 1 type of instrument. Retain the unique id, and the name for each musician, their salary as well as their level of expertise for the type of instrument they play. For each type of instrument retain the type name (e.g. Violin, Trumpet etc.) and the instrument category (e.g. strings, wind, brass etc.). All instrument types will have at least 1 assigned musician.
- e) An art researcher has asked you to design a database to record the details of world-famous museums and the artists and their paintings that are displayed in these museums. For each painting, the researcher wants to know the year painted, title, and painting genre. The unique id and the date of birth of each artist must be recorded as well as their name and nationality. For each museum, record the details of its unique id, its name, city and specialty, if it has one. Each museum assigns its own unique catalog number for each painting in its collection.

Submission: Take a screen shot of the models. Paste the screen shots in a MS Word document. Submit the MS Word document to D2L.

You need to write your name and student number on top of the page.